

Student Grade Tracker - Project Reflection

Module: Programming 1 (Week 7 Formative Project)

Language: Python 3

URL: <https://github.com/sanagbah-007/ProgrammingOneFormativeProject.git>

What I Learnt

Through this project, I gained practical experience applying inheritance-based programming concepts in Python:

- **Inheritance & Reusability:** I learned how to create a parent class (Assignment) for shared attributes such as title, score, and date, and to extend it with child classes (Homework and Exam) using `super().__init__()`. This saved time and prevented code repetition.
- **Separation of Concerns:** Dividing the project into `models.py` (data structures), `tracker.py` (data management), and `main.py` (user interface) made the code much easier to read, test, and debug.
- **Defensive Programming:** I learned how to write validation loops and input handler classes to catch user errors (such as empty strings or letters entered instead of numbers) before they cause a crash.

Challenges Faced

1. **Attribute Naming Across Files:** One challenge was keeping attribute names consistent across files (for example, ensuring `assignment_type` in `models.py` matched the filtering logic in `tracker.py`). Careful testing helped identify and fix these mismatches quickly.
2. **Avoiding Zero-Division Errors:** Calculating percentage averages when no assignments were present (or if a maximum score was 0) initially risked crashing the program. I resolved this by adding checks that return 0.0% if total maximum scores equal zero.
3. **Data Formatting in the CLI:** Making sure all table columns lined up cleanly regardless of word lengths required learning and using string padding tricks like `<12`.

Future Improvements

Given more development time, I would like to improve the project by:

- **Adding Data Persistence:** Saving assignment records to a local CSV file so student data stays saved even after closing the terminal session.
- **Low Score Alerts:** Automatically highlighting any assignment score below a certain percentage threshold (e.g., under 50%) to alert the student.
- **Unit Tests:** Adding automated test functions using Python's built-in `unittest` module to verify calculations automatically.